



Crash Course in Maths for Cognitive Science

Session 1 | Probability

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What today is about

Almost everything we study in cognitive science is *uncertain*: the next response of a participant, the number of spikes a neuron fires, whether a model gets an answer right.

Probability is simply the careful mathematics of that uncertainty. Today we build it from the ground up.

- Describe what can happen: *sample spaces* and *events*.
- Combine chances with a few simple *rules*.
- Update a belief when new evidence arrives: **Bayes' theorem**.
- Meet the handful of *distributions* that describe most real data.

Throughout, we stop and *solve examples together*; the ideas only stick once you use them.

The one idea to hold onto

Everything today leads to one destination.

If you remember only one thing

Bayes' theorem tells us how observing evidence should change what we believe. It is the mathematical backbone of Bayesian models of perception, learning, and decision-making, and of machine learning.

We will spend the first hour building the tools we need to understand it, and the second hour using it and meeting the common distributions.

A probability is just a number between 0 and 1

Before any formulas, fix the picture in your head.

- A probability measures our *degree of certainty* that something happens.
- 0 means impossible, 1 means certain, 0.5 means “as likely as not”.
- Everything else is rules for combining these numbers without contradicting ourselves.

Intuition

Think of one unit of “belief” that we spread across all the things that could happen. Probability is just how much of that belief sits on each outcome.

Why this matters for cognitive science

This is not only mathematics; it is a theory of the mind.

A leading view in modern cognitive science is that the brain is, in part, a *probabilistic inference machine*. Perception is treated as inferring the most probable cause of ambiguous sensory data; learning as updating beliefs from experience.

Link to cognitive science

Bayesian models of cognition, ideal-observer analysis, signal-detection theory in psychophysics, and the probabilistic models underlying machine learning all speak the language we build today. Learning it lets you read a large part of the modern literature.

Step 1: list everything that can happen

To reason about chance we first say, precisely, what the possible outcomes are.

Definition

The **sample space** Ω is the set of all possible outcomes of an experiment. An **event** A is a subset of Ω , a collection of outcomes we care about.

A few concrete examples:

- One die roll: $\Omega = \{1, 2, 3, 4, 5, 6\}$; the event “even” is $\{2, 4, 6\}$.
- A detection task: $\Omega = \{\text{yes}, \text{no}\}$.
- Spikes in one second: $\Omega = \{0, 1, 2, 3, \dots\}$.

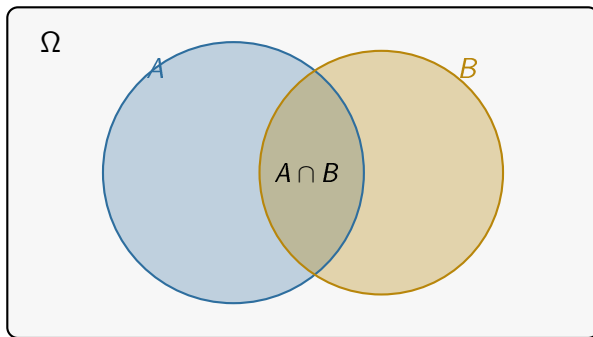
Events are yes/no questions about the outcome

An event is just a question with a yes/no answer: “was the die even?” splits Ω into the outcomes where the answer is yes and those where it is no.

Because events are sets, we combine them with the same words we use in ordinary logic:

- $A \cup B$, “*A or B*”: outcomes in either (or both).
- $A \cap B$, “*A and B*”: outcomes in both at once.
- A^c , “*not A*”: every outcome outside A .
- A, B are *disjoint* if they cannot both happen: $A \cap B = \emptyset$.

Picturing events: the Venn diagram



The whole rectangle is Ω . The overlap is “ A and B ”; the two circles together are “ A or B ”; everything outside both circles is “neither”.

When outcomes are equally likely, we just count

If every outcome is equally likely, probability becomes a counting problem:

$$\mathbb{P}(A) = \frac{\# \text{ outcomes in } A}{\# \text{ outcomes in } \Omega}.$$

So the skill we need is counting outcomes efficiently.

Permutations vs. combinations

Order matters (permutations): $\frac{n!}{(n-k)!}$ ways to arrange k of n . **Order does not**

(combinations): $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ ways to choose k of n .

Three counting facts worth knowing

Useful fact

1. **Multiplication principle:** a independent choices then b more give $a \times b$ possibilities.
2. $\binom{n}{k} = \binom{n}{n-k}$, choosing who is *in* is the same as choosing who is *out*.
3. $\binom{n}{0} = \binom{n}{n} = 1$, $\binom{n}{1} = n$.

Intuition

The multiplication principle is why a task with many independent slots explodes so fast, 5 yes/no items already give $2^5 = 32$ response patterns.

Worked example 1: designing a little experiment

Example

A stimulus set has 8 images.

1. How many different *sets* of 3 images could you show (order on the screen does not matter)?
2. How many different *ordered sequences* of 3 images could you present?

Try it yourself first — the solution is on the next slide.

Worked example 1: solution

1. A set ignores order, so use a combination: $\binom{8}{3} = \frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} = 56$.
2. A sequence cares about order, so use a permutation: $8 \cdot 7 \cdot 6 = 336$.

Notice $336 = 56 \times 3!$: each set of 3 images can be shown in $3! = 6$ orders, so the sequence count is the set count times 6. That relationship (permutations = combinations $\times k!$) is worth remembering.

Worked example 2: how many response patterns?

Example

A short questionnaire has 5 yes/no items, and separately a reaction-time trial can end in one of 3 ways (correct, error, or no-response).

1. How many distinct patterns of answers to the 5 items are possible?
2. Including the trial outcome, how many combined patterns are there?

Try it yourself first — the solution is on the next slide.

Worked example 2: solution

1. Each item has 2 possibilities and they are independent, so by the multiplication principle $2^5 = 32$ patterns.
2. Attach the 3 trial outcomes: $32 \times 3 = 96$ combined patterns.

Intuition

This is why the space of possible “behaviours” grows so quickly with the number of measurements, and why we need probability (not enumeration) to reason about it.

Practice: counting

Try these on your own; we will solve them together on the board.

Practice — try it

1. From a group of 10 participants, how many ways are there to form a 4-person focus group?
2. In how many different orders can 6 pictures be shown to a subject?
3. A PIN uses 4 digits (0–9), repeats allowed. How many PINs exist?

Answers. (1) $\binom{10}{4} = 210$. (2) $6! = 720$. (3) $10^4 = 10\,000$.

Three common-sense rules (the axioms)

Any sensible way of assigning probabilities must obey three rules. They simply formalise common sense.

Kolmogorov's axioms

- (1) $\mathbb{P}(A) \geq 0$, no negative chances.
- (2) $\mathbb{P}(\Omega) = 1$, something must happen.
- (3) If A and B are disjoint, $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B)$.

Everything else in probability can be derived from just these three statements.

Rules that follow from the axioms

Useful fact

Complement: $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.

Addition: $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

Inclusion–exclusion (three events): $\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C)$.

Intuition

Why subtract? Adding $\mathbb{P}(A) + \mathbb{P}(B)$ counts the overlap $A \cap B$ *twice*, so we remove it once. For three sets, the “+ triple” at the end puts back the piece we removed one time too many.

Worked example 3: overlapping groups

Example

In a cohort, 70% passed a maths exam (M), 50% passed a statistics exam (S), and 40% passed *both*.

1. What fraction passed *at least one* of the two?
2. What fraction passed *neither*?
3. What fraction passed *exactly one*?

Try it yourself first — the solution is on the next slide.

Worked example 3: solution

1. $\mathbb{P}(M \cup S) = 0.70 + 0.50 - 0.40 = 0.80.$
2. $\mathbb{P}(\text{neither}) = 1 - \mathbb{P}(M \cup S) = 1 - 0.80 = 0.20.$
3. $\mathbb{P}(\text{exactly one}) = \mathbb{P}(M \cup S) - \mathbb{P}(M \cap S) = 0.80 - 0.40 = 0.40.$

A quick sanity check: “both” (0.40) + “exactly one” (0.40) + “neither” (0.20) = 1, as it must.

Worked example 4: the “at least one” trick

Example

In a rapid stream of 3 independent frames, a faint target appears on each frame with probability 0.2. What is the probability of seeing *at least one* target across the three frames?

Intuition

“At least one” is awkward head-on (exactly one? exactly two? all three?). The complement, “*none*”, is a single, easy calculation.

Try it yourself first — the solution is on the next slide.

Worked example 4: solution

The chance of *no* target on one frame is $1 - 0.2 = 0.8$. Across three independent frames,

$$\mathbb{P}(\text{no target at all}) = 0.8^3 = 0.512,$$

so

$$\mathbb{P}(\text{at least one}) = 1 - 0.512 = 0.488.$$

Common pitfall

Do not add $0.2 + 0.2 + 0.2 = 0.6$, that double-counts the frames where more than one target appears, and would even exceed 1 for enough frames. Always route “at least one” through the complement.

Practice: probability rules

Practice — try it

In a sample, $\mathbb{P}(\text{coffee}) = 0.5$, $\mathbb{P}(\text{tea}) = 0.3$, and $\mathbb{P}(\text{both}) = 0.1$. Find:

1. $\mathbb{P}(\text{coffee or tea})$;
2. $\mathbb{P}(\text{neither})$;
3. $\mathbb{P}(\text{exactly one of the two})$.

Answers. (1) 0.7. (2) 0.3. (3) 0.6.

Learning from information: conditional probability

The central question of the whole session: *how does knowing one thing change the probability of another?*

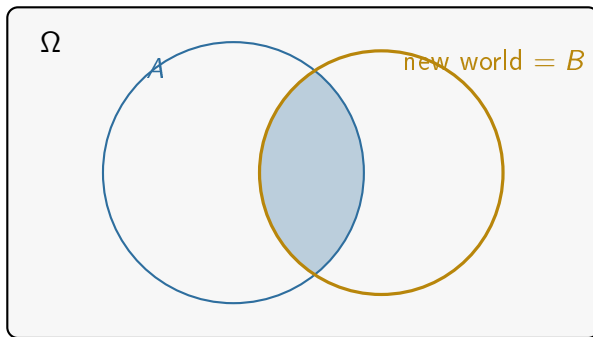
Definition

The **conditional probability** of A given B (with $\mathbb{P}(B) > 0$) is

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Read it as: “among the cases where B happened, how often does A also happen?”

Conditioning means zooming in



Conditioning on B throws away everything outside the B circle. Our new “world” is B , and $\mathbb{P}(A \mid B)$ is the fraction of that world covered by A . Dividing by $\mathbb{P}(B)$ rescales it so the shrunken world again totals 1.

Worked example 5: conditioning changes the odds

Example

Roll a fair die.

1. What is $\mathbb{P}(\text{even})$?
2. You are told the result is *greater than* 3. Now what is $\mathbb{P}(\text{even} \mid > 3)$?

Try it yourself first — the solution is on the next slide.

Worked example 5: solution

1. Even numbers are $\{2, 4, 6\}$, so $\mathbb{P}(\text{even}) = \frac{3}{6} = \frac{1}{2}$.
2. “Greater than 3” leaves the world $\{4, 5, 6\}$. Among those, the even ones are $\{4, 6\}$, so

$$\mathbb{P}(\text{even} \mid > 3) = \frac{2}{3}.$$

Information moved the probability from $\frac{1}{2}$ up to $\frac{2}{3}$, that is exactly what “learning” means, quantified.

Worked example 6: a signal-detection table

Link to cognitive science

In psychophysics, a subject says “yes, I saw it” or “no”. Crossing that with whether a stimulus was truly present gives the four cells of *signal-detection theory*: hits, misses, false alarms, correct rejections.

Example

Out of 200 trials: of the 100 trials where a signal was *present*, the subject said “yes” on 80; of the 100 *absent* trials, the subject said “yes” on 30.

1. Find the *hit rate* $\mathbb{P}(\text{“yes”} \mid \text{present})$.
2. Find the *false-alarm rate* $\mathbb{P}(\text{“yes”} \mid \text{absent})$.

Try it yourself first — the solution is on the next slide.

Worked example 6: solution

$$\text{hit rate} = \mathbb{P}(\text{"yes"} \mid \text{present}) = \frac{80}{100} = 0.80,$$

$$\text{false-alarm rate} = \mathbb{P}(\text{"yes"} \mid \text{absent}) = \frac{30}{100} = 0.30.$$

Intuition

These two conditional probabilities are the coordinates of a point on the *ROC curve*. A subject who is more *sensitive* pushes the hit rate up without raising the false-alarm rate; a change of *criterion* (being more willing to say “yes”) slides both up together. Separating sensitivity from bias is the whole point of signal-detection theory, and it is built entirely from conditional probabilities.

The multiplication rule

Rearranging the definition gives a way to build the probability of “ A and B ” step by step:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A | B) \mathbb{P}(B) = \mathbb{P}(B | A) \mathbb{P}(A).$$

Intuition

Read left to right: to get both, first B happens (probability $\mathbb{P}(B)$), and *then*, in that world, A happens (probability $\mathbb{P}(A | B)$). Chaining these is how we follow a sequence of dependent steps.

Averaging over scenarios: total probability

Often we know how likely A is *within* each of several scenarios, and want its overall probability.

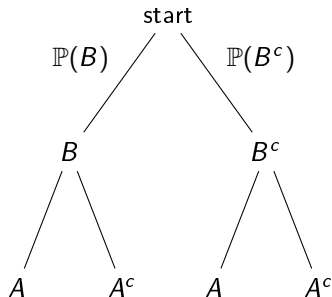
Law of total probability

If $B_1, \dots, B_k$ are disjoint and cover everything, then

$$\mathbb{P}(A) = \sum_i \mathbb{P}(A | B_i) \mathbb{P}(B_i).$$

In words: work out how likely A is in each scenario, then take a weighted average, weighting each scenario by how likely it is.

The probability tree



Multiply probabilities along a branch, then *add* across the branches that end in A :

$$\mathbb{P}(A) = \mathbb{P}(B)\mathbb{P}(A \mid B) + \mathbb{P}(B^c)\mathbb{P}(A \mid B^c).$$

Worked example 7: overall accuracy across trial types

Example

An experiment mixes two kinds of trial. *Easy* trials are 60% of the session and are answered correctly with probability 0.9; *hard* trials are the other 40% and are answered correctly with probability 0.6.

If a trial is picked at random, what is the overall probability of a correct response?

Try it yourself first — the solution is on the next slide.

Worked example 7: solution

Split into the two scenarios and take the weighted average:

$$\mathbb{P}(\text{correct}) = \underbrace{0.6 \times 0.9}_{\text{easy}} + \underbrace{0.4 \times 0.6}_{\text{hard}} = 0.54 + 0.24 = 0.78.$$

Intuition

Overall accuracy is *not* the simple average of 0.9 and 0.6 ($= 0.75$): the easy trials are more frequent, so they pull the average up. Whenever you pool across conditions, remember the weights, this is also the seed of *Simpson's paradox* in Session 2.

Worked example 8: which box did it come from?

Example

Box 1 holds 3 red and 1 blue token; Box 2 holds 1 red and 3 blue. You pick a box at random ($\frac{1}{2}$ each) and draw one token.

1. What is $\mathbb{P}(\text{red})$?
2. (*Look ahead.*) If the token is red, does that make Box 1 or Box 2 more likely?

Try it yourself first — the solution is on the next slide.

Worked example 8: solution

$$\mathbb{P}(\text{red}) = \underbrace{\frac{3}{4} \cdot \frac{1}{2}}_{\text{Box 1}} + \underbrace{\frac{1}{4} \cdot \frac{1}{2}}_{\text{Box 2}} = \frac{3}{8} + \frac{1}{8} = \frac{1}{2}.$$

A red token is more likely to have come from Box 1 (which is red-rich). To turn that intuition into a number we must *reverse* the conditioning, exactly what Bayes' theorem does, using this $\frac{1}{2}$ as its denominator.

Independence: when information tells you nothing

Definition

A and B are **independent** if knowing one tells you nothing about the other:

$$\mathbb{P}(A | B) = \mathbb{P}(A), \quad \text{equivalently} \quad \mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

Independence is what lets us multiply probabilities of separate trials, the coin does not remember its last flip. It is an assumption we make constantly, and it is usually an *approximation*.

Independent is not the same as disjoint

Common pitfall

Independent and *disjoint* are almost opposites. Disjoint events cannot both happen, so learning that one occurred tells you the other did *not*, that is maximal *dependence*. Independence means the events carry *no* information about each other.

If in doubt, go back to the definition: independence is about $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, not about the events being mutually exclusive.

Worked example 9: are two neurons independent?

Example

On any trial, neuron A fires with probability 0.4 and neuron B with probability 0.3.

1. If they fired independently, what would $\mathbb{P}(\text{both fire})$ be?
2. You actually measure $\mathbb{P}(\text{both fire}) = 0.20$. Are they independent? What does the discrepancy suggest?

Try it yourself first — the solution is on the next slide.

Worked example 9: solution

1. Under independence, $\mathbb{P}(\text{both}) = 0.4 \times 0.3 = 0.12$.
2. The measured 0.20 is well above 0.12, so the neurons are **not** independent, they fire together more often than chance.

Link to cognitive science

That excess is a *noise correlation*. Whether neurons in a population fire independently or in a correlated way changes how much information the population carries, a central question in systems neuroscience. Many models *assume* independence for tractability; measuring the gap tells you how wrong that assumption is.

Bayes' theorem: updating a belief

Now the destination. Bayes' theorem flips a conditional probability around, which is exactly what “learning from evidence” requires.

Key idea

$$\mathbb{P}(H | E) = \frac{\mathbb{P}(E | H) \mathbb{P}(H)}{\mathbb{P}(E)} \propto \mathbb{P}(E | H) \mathbb{P}(H),$$

where H is a hypothesis and E is the evidence we observe.

The four pieces of Bayes' theorem

- **Prior** $\mathbb{P}(H)$, what we believed *before* the evidence.
- **Likelihood** $\mathbb{P}(E \mid H)$, how well the hypothesis *predicts* the evidence.
- **Posterior** $\mathbb{P}(H \mid E)$, our *updated* belief.
- **Evidence** $\mathbb{P}(E)$, a normaliser, from the law of total probability:
$$\mathbb{P}(E) = \mathbb{P}(E \mid H)\mathbb{P}(H) + \mathbb{P}(E \mid H^c)\mathbb{P}(H^c).$$

Intuition

New belief $\propto$ (how well the hypothesis predicted what we saw) $\times$ (how much we believed it beforehand).

Worked example 10: the famous positive test

Example

A disease affects 1% of people. A test correctly flags 99% of true cases (sensitivity) but also wrongly flags 5% of healthy people (false-positive rate).

A randomly chosen person tests *positive*. What is the probability they actually have the disease, roughly 99%, about 50%, or much less?

Try it yourself first — the solution is on the next slide.

Worked example 10: solution (with Bayes)

Write D = has the disease, $+$ = positive test. The overall chance of a positive:

$$\mathbb{P}(+) = \underbrace{0.99 \times 0.01}_{\text{sick}=0.0099} + \underbrace{0.05 \times 0.99}_{\text{healthy}=0.0495} = 0.0594.$$

$$\mathbb{P}(D | +) = \frac{0.99 \times 0.01}{0.0594} = \frac{0.0099}{0.0594} = \frac{1}{6} \approx 16.7\%.$$

Far below the 99% that intuition suggests.

Worked example 10: the same answer, in plain counts

Imagine 10,000 people.

- 100 are sick; the test catches 99 of them.
- 9,900 are healthy; 5% are wrongly flagged = 495.
- So $99 + 495 = 594$ test positive, but only 99 are truly sick: $\frac{99}{594} = \frac{1}{6}$.

Link to cognitive science

People, including clinicians, badly overestimate this, focusing on the test's accuracy and neglecting the low *base rate*. This is the **base-rate fallacy** (Kahneman & Tversky), and its courtroom cousin is the “prosecutor’s fallacy”. Natural frequencies, as here, largely cure it.

The Bayesian brain, in one line

Link to cognitive science

The very same equation is the leading formal account of perception: the brain combines a **prior** (what is usually out there) with a **likelihood** (what the noisy senses report) to form a **posterior** (what it perceives). Illusions are then not failures but sensible inferences under a misleading prior.

So base-rate neglect and the “Bayesian brain” are two faces of the one theorem, one where people violate it, one where the brain seems to obey it.

A shortcut: Bayes in odds form

Sometimes it is quicker to work with *odds* and skip the denominator entirely.

Useful fact

$$\underbrace{\frac{\mathbb{P}(H | E)}{\mathbb{P}(H^c | E)}}_{\text{posterior odds}} = \underbrace{\frac{\mathbb{P}(E | H)}{\mathbb{P}(E | H^c)}}_{\text{likelihood ratio}} \times \underbrace{\frac{\mathbb{P}(H)}{\mathbb{P}(H^c)}}_{\text{prior odds}}.$$

Intuition

Evidence *multiplies* the odds. Independent pieces of evidence multiply their likelihood ratios, so in log-odds, evidence simply *adds up*. That running total is exactly what accumulator / drift-diffusion models of decision-making compute, and what neurons in area LIP appear to track.

Worked example 11: reading a neural detector

Example

A neural “detector” fires on 90% of trials when a target is present, and on 20% of trials when it is absent. Targets are rare: they occur on only 10% of trials.

The detector fires. What is the probability the target was actually present?

Try it yourself first — the solution is on the next slide.

Worked example 11: solution

Let T = target present, F = detector fires.

$$\mathbb{P}(F) = \underbrace{0.9 \times 0.1}_{\text{present}=0.09} + \underbrace{0.2 \times 0.9}_{\text{absent}=0.18} = 0.27,$$

$$\mathbb{P}(T | F) = \frac{0.9 \times 0.1}{0.27} = \frac{0.09}{0.27} = \frac{1}{3}.$$

Even a fairly reliable detector, applied to a *rare* target, is right only a third of the time when it fires, the base rate again. *Two* independent fires, though, would multiply the odds and push the posterior much higher.

Practice — try it

A jar of candies is either *ordinary* (30% red) or *special* (70% red); beforehand each is equally likely. You draw one candy and it is red.

1. What is $\mathbb{P}(\text{red})$ overall?
2. What is $\mathbb{P}(\text{special} \mid \text{red})$?
3. Redo part (2) using prior odds $\times$ likelihood ratio.

Answers. (1) $\mathbb{P}(\text{red}) = 0.5$. (2) 0.7. (3) odds $1:1 \times \frac{0.7}{0.3} = \frac{7}{3} \Rightarrow 0.7$.

From outcomes to numbers: random variables

So far outcomes have been labels (heads, even, positive). Usually we want *numbers* we can average and plot.

Definition

A **random variable** X assigns a number to every outcome. It is *discrete* if it takes separate values $(0, 1, 2, \dots)$ and *continuous* if it can take any value in a range.

Examples: the number of correct trials out of 20 (discrete); a reaction time in milliseconds (continuous); a spike count (discrete); a voltage (continuous).

Where a random variable's probability “lives”

Useful fact

Discrete, the *probability mass function* $p(x) = \mathbb{P}(X = x)$; the masses are ≥ 0 and sum to 1.

Continuous, the *probability density function* $f(x) \geq 0$; areas under it give probabilities, and the total area is 1.

Either, the *cumulative distribution function* $F(x) = \mathbb{P}(X \leq x)$, rising from 0 to 1.

Common pitfall

A density is *not* a probability: $f(x)$ can exceed 1, and $\mathbb{P}(X = x) = 0$ for a single continuous value. Only *intervals* (areas) carry probability.

Worked example 12: completing a distribution

Example

A subject's number of errors in a block, $X \in \{0, 1, 2, 3\}$, has

$$p(0) = 0.1, \quad p(1) = 0.3, \quad p(2) = 0.4, \quad p(3) = ?$$

1. Find $p(3)$.
2. Find $\mathbb{P}(X \geq 2)$.

Try it yourself first — the solution is on the next slide.

Worked example 12: solution

1. The masses must sum to 1: $p(3) = 1 - (0.1 + 0.3 + 0.4) = 0.2$.
2. $\mathbb{P}(X \geq 2) = p(2) + p(3) = 0.4 + 0.2 = 0.6$.

Always use “the masses sum to 1” as both a way to find a missing value and a check that a proposed distribution is legal.

Four distributions cover most of what you will meet

A “distribution” is a named, reusable shape of randomness. We will meet four, each with its own story, learn the story, not just the formula.

- **Bernoulli**, a single yes/no trial.
- **Binomial**, counting successes across many trials.
- **Poisson**, counts of rare events (e.g. spikes).
- **Gaussian**, the bell curve of noise and averages.

Bernoulli and Binomial

Bernoulli(p) is one yes/no trial: success with probability p . It is the atom of all binary data, one coin flip, one correct/incorrect response.

$$\mathbb{E}[X] = p, \quad \text{Var}(X) = p(1 - p).$$

Binomial(n, p) counts the successes in n independent Bernoulli trials:

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad \mathbb{E}[X] = np, \quad \text{Var}(X) = np(1 - p).$$

So a Binomial is just a *sum* of n Bernoullis.

Two useful relationships

Useful fact

- **Rare events:** a Binomial with large n and small p , with $np = \lambda$ moderate, is well approximated by a **Poisson**(λ).
- **Many trials:** a Binomial with large n is well approximated by a **Normal** with mean np and variance $np(1 - p)$ (a preview of the Central Limit Theorem).

Intuition

The same “count of successes” becomes Poisson when successes are rare, and bell-shaped when they are many, which is why these four distributions are so tightly connected.

Worked example 13: a two-alternative task

Link to cognitive science

In a two-alternative forced-choice (2AFC) task, each trial is right or wrong, and chance is 50%. Accuracy over n trials is Binomial.

Example

A participant performs at 75% accuracy over $n = 12$ independent trials. Let X be the number correct.

1. What is $\mathbb{E}[X]$?
2. What is $\mathbb{P}(X = 12)$ (a perfect block)?

Try it yourself first — the solution is on the next slide.

Worked example 13: solution

1. $\mathbb{E}[X] = np = 12 \times 0.75 = 9$ correct on average.
2. $\mathbb{P}(X = 12) = 0.75^{12} \approx 0.0317$, about a 3% chance of a perfect block, even at 75% accuracy.

Intuition

Plotting “proportion correct” against stimulus strength traces the *psychometric function*; its steepness reflects sensitivity, and where it crosses a threshold defines the *just-noticeable difference*. The Binomial is the noise model underneath every point on that curve.

Poisson: counts of rare events

When events happen independently at a steady average rate, the number in a fixed window follows a **Poisson**(λ), where λ is *both* the mean and the variance:

$$\mathbb{P}(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad \mathbb{E}[X] = \text{Var}(X) = \lambda.$$

Link to cognitive science

This is the standard first model of **neural spike counts**. The ratio variance/mean is the *Fano factor*; it equals 1 for a pure Poisson process, and departures from 1 reveal extra structure (bursting, or regularity) in real spike trains.

Worked example 14: silent windows

Example

A neuron fires as a Poisson process with mean $\lambda = 4$ spikes per 100 ms window.

1. What is the probability of observing *no* spikes in a window?
2. What is the variance of the spike count?

Try it yourself first — the solution is on the next slide.

Worked example 14: solution

1. $\mathbb{P}(X = 0) = \frac{4^0 e^{-4}}{0!} = e^{-4} \approx 0.018$, silent windows happen about 1.8% of the time.
2. For a Poisson, $\text{Var}(X) = \lambda = 4$ (so the standard deviation is 2).

Because mean = variance, spike counts are intrinsically noisy: at a mean of 4, counts of 2 or 6 are entirely ordinary. This “Poisson variability” is a baseline every neural model must reckon with.

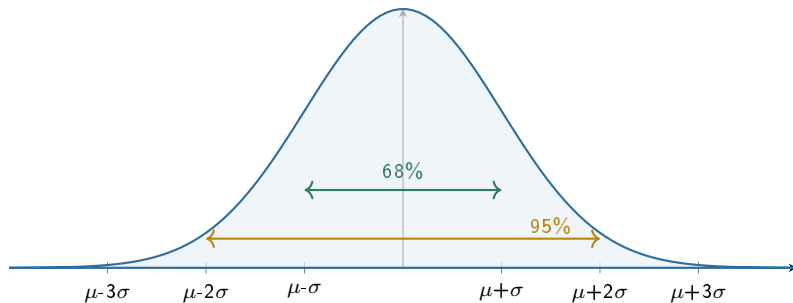
The Gaussian: the bell curve

The **Normal** (Gaussian) distribution $\mathcal{N}(\mu, \sigma^2)$ is the continuous workhorse. It is symmetric and bell-shaped, centred at the mean μ , with its width set by the standard deviation σ :

$$f(x) = \frac{1}{\sqrt{2\pi} \sigma} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

The next slide shows the one fact that lets you reason about it without any integral.

The 68–95–99.7 rule



About 68% of the probability lies within $\pm 1\sigma$, 95% within $\pm 2\sigma$, and 99.7% within $\pm 3\sigma$, for every Normal.

Reading off Normal probabilities: the z-score

To use the rule for any Normal, *standardise*: measure how many standard deviations a value is from the mean,

$$z = \frac{x - \mu}{\sigma}.$$

Intuition

The z-score strips away the units and the scale, leaving a pure “how unusual is this?” number. It is what lets us compare a reaction time with an IQ score, and it is the currency of the 68–95–99.7 rule.

Worked example 15: how unusual is this reaction time?

Example

Reaction times in a task are approximately $\mathcal{N}(\mu = 600, \sigma = 80)$ ms.

1. What is the z-score of a 760 ms response?
2. Roughly what fraction of responses are *slower* than 760 ms?

Try it yourself first — the solution is on the next slide.

Worked example 15: solution

1. $z = \frac{760 - 600}{80} = 2.0$: this response is 2 standard deviations above the mean.
2. Since 95% of the mass lies within $\pm 2\sigma$, the two tails together hold 5%, so the upper tail is about $\frac{5\%}{2} = 2.5\%$.

Link to cognitive science

A caveat worth stating: real reaction-time distributions are *right-skewed*, a Gaussian plus a long exponential tail (the “ex-Gaussian”). The Normal is a good first model near the centre, but the slow tail carries real cognitive signal (lapses, attentional dips).

The four distributions at a glance

Distribution	Models	Support	Mean	Variance
Bernoulli(p)	one yes/no trial	$\{0, 1\}$	p	$p(1 - p)$
Binomial(n, p)	# successes in n	$\{0, \dots, n\}$	np	$np(1 - p)$
Poisson(λ)	counts of rare events	$\{0, 1, 2, \dots\}$	λ	λ
Normal(μ, σ^2)	noise / averages	$\mathbb{R}$	μ	σ^2

A Binomial is a sum of Bernoullis; for rare successes it becomes Poisson, and for many trials it becomes Normal.

Why the bell curve is everywhere

Link to cognitive science

Whenever a quantity is the sum of many small, independent influences, its distribution is approximately Normal, regardless of the details. This is the **Central Limit Theorem**, which we make intuitive next session.

It explains why test scores, measurement errors, and averaged neural signals all look bell-shaped, and why Gaussian noise is the default assumption in almost every model of perception and data analysis.

Practice — try it

1. A subject responds correctly on 70% of trials, independently. Over 8 trials, what is the expected number correct, and $\mathbb{P}(\text{all 8 correct})$?
2. A neuron fires at $\lambda = 2$ spikes per window. What is $\mathbb{P}(X = 0)$?
3. Reaction times are $\mathcal{N}(500, 50^2)$ ms. Roughly what fraction fall between 450 and 550 ms?

Answers. (1) mean 5.6, $\mathbb{P}(\text{all 8}) = 0.7^8 \approx 0.058$. (2) $e^{-2} \approx 0.135$. (3) $\approx 68\%$ (within $\pm 1\sigma$).

Session 1: what we built

- Probabilities are numbers in $[0, 1]$ obeying the complement and addition rules.
- **Conditional probability** formalises learning from information (hit and false-alarm rates, total probability); *independence* is its absence.
- **Bayes' theorem** turns evidence into updated belief, base-rate neglect, the Bayesian brain, and evidence accumulation.
- **Random variables** and four *distributions* describe binary trials, counts, rare events, and bell-shaped noise.

Next session: from single quantities to *data*, summarising it and inferring from it.